

Adham Aly

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EDUCATION

University of Toronto

Honours Bachelor of Science, Computer Science Specialist (AI Focus)

Toronto, ON

Sept. 2026 – 2030 (Expected)

PROJECTS

WMOJ — Competitive Programming Online Judge | TypeScript, Next.js, PostgreSQL, Docker

- Co-founded WMOJ, an open-source competitive-programming judge with 80 problems, timed contests, and live leaderboards, used by 3+ Halton District School Board schools and running the WOSS Dual Olympiad and TriOlympiad
- Built the grading server that compiles and runs untrusted Python and C++ code inside an nsjail sandbox, with a custom seccomp filter that blocks network access and process tampering, plus CPU, memory, and output limits
- Found and fixed a bug that had graded 94 real timeouts as wrong answers, writing a small C program that records each run's exact CPU time and memory use
- Wrote 190+ unit tests and 39 recorded end-to-end grading scenarios that CI replays against the real Docker image on every push and nightly
- Ran a security review of the web app that fixed 70 defects, adding 64 PostgreSQL row-level security policies and closing holes that exposed answer keys and student code and let anyone forge submissions

Scire — Peer-to-Peer Tutoring LMS | Next.js, React, TypeScript

- Built Scire, a full-stack tutoring learning-management system in Next.js, React, and TypeScript, with tutor-tutee matching and scheduling, session tracking, and admin dashboards
- Deployed it into active use at a high-school tutoring club to coordinate and track sessions; currently pursuing Halton District School Board approval to expand district-wide

EXPERIENCE

President, Computer Science Club

White Oaks Secondary School

Sept. 2024 – July 2026

Oakville, ON

- Directed a 15+ person executive team running computer-science and competitive-programming events, including the WOSS Dual Olympiad, owning logistics, marketing, and sponsorship outreach
- Taught weekly data-structures, algorithms, and software-engineering lessons to a class of 50+ students, building their competitive-programming and problem-solving skills
- Led a 10-person curriculum subteam authoring lesson plans, in-class activities, and original competitive-programming contest problems used across sessions and competitions

Director, WOSS TriOlympiad

White Oaks Secondary School

Sept. 2024 – July 2026

Oakville, ON

- Directed the WOSS TriOlympiad, a free three-day physics, math, and computer-science competition with solo and team rounds, drawing 100+ interested students from across the Halton District School Board
- Coordinated a 16-person organizing team across the school's Math, Physics, and Computer Science clubs to write original contest problems and run the event, and led sponsorship outreach with tiered packages from \$100 to \$1,500
- Co-built the event website (triolympiad.ca) in Next.js and TypeScript, with online registration that saved sign-ups to Google Sheets and sent automatic confirmation emails

TECHNICAL SKILLS

Languages: C, C++, TypeScript, JavaScript, Python, SQL, x86 Assembly, HTML/CSS

Frameworks & Libraries: Next.js, React, Node.js

Developer Tools: Docker, Git, Linux, QEMU, GCC, Make/CMake

Concepts: Data Structures & Algorithms, Operating Systems, Machine Learning & LLMs, REST APIs, Containerization